

Semantic Interpretability in Deep Knowledge Tracing via Representational Grounding: Toward Individualized Learning Trajectories

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Abstract

Knowledge Tracing, which enables the estimation of how students' knowledge evolves as they interact with educational content, is a cornerstone of intelligent tutoring systems. Traditional models coexist with new deep learning approaches that deliver better predictions but suffer from a critical drawback: lack of interpretability, which is particularly problematic in educational contexts. This study introduces iDKT, an interpretable-by-design Deep Knowledge Tracing model that utilizes *Representational Grounding* to ground latent representations to educational constructs defined by traditional models that are intrinsically interpretable. We introduce a formal validation framework to verify Representational Grounding and, using Bayesian Knowledge Tracing as a reference, evaluate iDKT across multiple educational datasets. Results demonstrate that iDKT maintains state-of-the-art predictive performance while yielding additional interpretable estimations at a significantly higher granularity than population-level bayesian approaches. Specifically, iDKT identifies student-level initial knowledge and learning velocities, providing mastery estimations that are more sensitive to the nuances of individual behavioral patterns than standard bayesian predictions. By anchoring internal representations to semantic concepts defined by the reference model, iDKT enables a pedagogically interpretable deep learning alternative and a practical tool for learner-centred systems through data-driven personalization.

Keywords: deep knowledge tracing; interpretability; representational grounding; bayesian knowledge tracing; transformer; educational data analysis; personalized learning

1. Introduction

1.1. Interpretability in Deep Knowledge Tracing

Knowledge Tracing (KT) constitutes a foundational problem in Artificial Intelligence in Education (AIED) and Intelligent Tutoring Systems (ITS). Fundamentally, it defines the task of modeling the dynamic evolution of a student's knowledge state over time based on their history of interactions with learning materials [1]. By estimating the probability that a student has mastered a specific skill at a given moment, KT systems enable the delivery of personalized instruction—recommending remedial content for struggling students or accelerating the curriculum for those who demonstrate proficiency. In an era where educational environments are becoming increasingly digital and diverse, the ability to accurately

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track and interpret student mastery is not merely a technical optimization but a critical requirement for effective, scalable, and learner-centred education.

Historically, the development of KT has been characterized by a dichotomy between two distinct paradigms [2]. The first, which predominated in the field for years, is characterized by probabilistic models such as Bayesian Knowledge Tracing [1] and Factor Analysis Models based on logistic regression—including the Additive Factor Model [3], Performance Factor Analysis (PFA) [4], and Knowledge Tracing Machine (KTM) [5]. Factor Analysis Models are theoretically supported by the Item Response Theory (IRT) [6], which has played a large role in educational assessment and measurement. These traditional approaches are intrinsically interpretable: they model learning as a process governed by explicit parameters related to pedagogical, cognitive or psychometric principles. Because the model structure mirrors a theoretical understanding of human learning, educators can inspect these values to validate the model's reasoning and trust its outputs. However, this transparency comes at the cost of representational rigidity; the simplifying assumptions required by these models often limit their predictive power, causing them to struggle with the complex, non-linear, and long-range dependencies inherent in real-world student behavior [7].

The second paradigm emerged with the advent of Deep Knowledge Tracing (DKT) [8], which shifted the focus to high-dimensional models. By leveraging Recurrent Neural Networks first and, more recently, Transformer architectures [9], DKT models treat student interaction histories as complex temporal sequences. These models have achieved state-of-the-art predictive performance, significantly outperforming classical baselines by automatically extracting latent features from vast amounts of data [2]. However, this substantial gain in predictive performance has frequently come at the cost of model transparency, creating a significant interpretability gap. Deep learning models are notoriously opaque “black boxes”, where the internal high-dimensional representations bear no direct correspondence to constructs that are interpretable in terms of human-understandable concepts or established domain knowledge. In a high-stakes domain such as education—where algorithmic decisions influence learning trajectories, grades, and opportunities—this lack of transparency poses critical challenges for practical adoption.

Interpretability in DKT remains an open issue, despite diverse attempts categorized by Bai et al. [10] into two primary modalities: *post-hoc* and *ante-hoc* approaches. Post-hoc methods attempt to explain a trained black-box model using techniques such as Layer-wise Relevance Propagation (LRP) [], perturbation analysis [], or local approximations like LIME []. However, as argued by Rudin [11], relying on post-hoc explanations for high-stakes decision-making is fundamentally flawed. These methods have limited utility for teachers and domain experts since they produce explanations related to the model's internal structures that are disconnected from pedagogically sound principles, thus requiring significant technical expertise to interpret.

The second category, *ante-hoc* methods, aims for intrinsic interpretability, encompassing two main approaches: traditional models that rely on simple, transparent structures, and DKT variants that achieve interpretability by incorporating additional modules or side information. Prominent examples in the latter group include Self-Attentive Knowledge Tracing (SAKT) [12], the first model to incorporate attention mechanisms, and Attentive Knowledge Tracing (AKT) [13], which achieved high predictive performance by integrating attention heads that apply different temporal distances to capture student interaction patterns with varying temporal resolution. However, it can be maintained that relying solely on attention mechanisms presents similar practical obstacles to post-hoc methods since they do not provide a clear interpretation of the model's predictions in terms of pedagogically sound principles.

Concerning the integration of side information into ante-hoc methods, Item Response Theory (IRT) [6] represents the most prominent theoretical foundation. IRT estimates the probability of a student correctly answering a question given their latent ability and the question's difficulty. This framework has been coupled with DKT models such as Deep-IRT [14], which leverages dynamic key-value memory networks to capture learners' trajectories; this allows for the inference of abilities and item difficulties via neural networks, which are subsequently operationalized to predict answer correctness. Similarly, TC-MIRT [15] embeds the parameters of a multidimensional IRT to predict students' states and generate interpretable parameters. Other approaches, such as KIKT [16], harness IRT to simulate learners' performance, while LANA [17] adopts a IRT variant to cluster students' ability levels. Finally, QIKT [18] features question-centric representations combined with an interpretable IRT layer. Beyond IRT, other researchers have leveraged diverse principles, such as constructive learning [19,20], learning curve and forgetting curve [21], finite state automaton (FSA) [22], classical test theory (CTT) [23], and monotonicity theory [24].

While providing insights into controlled scenarios, these approaches may struggle to encapsulate the multifaceted and dynamic nature of learning processes and the efficacy is often constrained by the specificity and scope of the underlying theories [10]. Moreover, the mere incorporation of principles from psychology or pedagogy does not guarantee interpretability. As noted by Von Rueden et al. [25], improved interpretability remains a secondary objective in most knowledge integration approaches used in machine learning, often overshadowed by performance optimization.

1.2. Semantic Interpretability

To achieve practical interpretability without compromising general applicability, we propose *Semantic Interpretability*—an approach that integrates the predictive strength of deep learning with the semantic advantages of traditional models. Under this framework, we define three essential criteria for model interpretability:

1. *Explicit Estimations*: The model produces numerical values for interpretable parameters that correspond to concepts acting as semantic anchors.
2. *Representational Grounding*: These estimations are grounded in the values provided for those concepts by an intrinsically interpretable reference model.
3. *Mathematical Derivation*: The model's final predictions are mathematically derived from these estimations, ensuring a transparent decision-making logic.

In the following section, we introduce iDKT (*Interpretable Deep Knowledge Tracing*), a novel Transformer-based architecture designed to satisfy the requirements of Semantic Interpretability and propose a *Representational Grounding* methodology to align iDKT estimations with educational constructs defined by a reference theory. To validate the approach, we define a rigorous metric for interpretability and analyze the trade-off between predictive performance and interpretability via a systematic *Pareto frontier* sweep. Our results demonstrate that iDKT achieves high theoretical alignment while preserving over 99% of the predictive capacity of unconstrained baselines. Notably, on the ASSISTments 2009 dataset, our experiments revealed that enhancing interpretability resulted in a performance gain, with an observed increase of +0.012 in AUC. These findings provide robust empirical support for the assertion by Rudin [11] that the inherent trade-off between predictive accuracy and model interpretability is often a myth, particularly in domains with significant underlying structure. This perspective is further supported by the *Rashomon Set* argument [26], which posits that for many real-world problems, the set of high-performing models is sufficiently large to contain solutions that are both accurate and intrinsically interpretable.

1.3. Individualized Learning Trajectories

For this study, we selected Bayesian Knowledge Tracing (BKT) as our anchoring theoretical framework. BKT models the learning process as a hidden Markov process governed by four explicit parameters: *Initial Mastery*, the probability a student knows a skill prior to practice; *Learn Rate*, the probability of transitioning from unlearned to learned states after a practice opportunity; *Guess*, the probability of a correct response despite a lack of mastery; and *Slip*, the probability of an incorrect response despite mastery. These parameters provide a causal narrative of student performance that is intuitive to educators and aligned with cognitive science principles.

Standard BKT suffers from a significant limitation: it typically operates at a population level, estimating a single set of parameters for all students interacting with a given skill. While *Individualized BKT* approaches exist, they are often computationally expensive and struggle with data sparsity [27]. The Representational Grounding used in the iDKT model offers a novel way to address individualization by leveraging the capabilities of Transformers to compress student interaction histories into a context vector. This allows the model to infer personalized initial knowledge and learning rates dynamically, transforming population-level BKT insights into high-granularity student diagnostics.

1.4. Research Hypotheses

The validation of our approach will be guided by three research hypotheses:

- **H1 (Representational Grounding):** Through Representational Grounding, the internal representations of an attention-based Transformer model can be aligned with pedagogically meaningful constructs to achieve Semantic Interpretability. We hypothesize that iDKT can achieve this kind of interpretability by exhibiting high linear recoverability with respect to BKT parameters across diverse educational datasets. Specifically, we posit that BKT constructs can be accurately recovered from iDKT's representations using simple linear transformations, which we validate through diagnostic probing with control tasks.
- **H2 (Predictive Performance-Interpretability Trade-off):** The pursuit of intrinsic interpretability does not necessarily imply a significant degradation in predictive performance. We can identify configurations that preserve near-optimal accuracy while achieving high semantic alignment. Furthermore, we posit that in certain contexts, the inductive bias introduced by pedagogical grounding can act as a regularizer, actually enhancing the model's predictive accuracy.
- **H3 (Estimations Granularity):** Besides the alignment with a reference model, Representational Grounding enables the discovery of fine-grained parameter values. Specifically, we hypothesize that iDKT provides superior diagnostic granularity by inferring personalized learning rates and initial mastery levels that capture behavioral heterogeneity more effectively than population-level probabilistic models.

The remainder of this paper is structured as follows. Section 2 describes the iDKT model, detailing the representational grounding mechanism and the multi-objective loss system. Section 5 presents the experimental setup. Section 6 discusses the empirical results, and Section 7 presents our final conclusions.

2. iDKT Model

We propose an Interpretable Deep Knowledge Tracing (iDKT) model with a Transformer-based architecture designed to bridge the gap between the high predictive capacity of deep learning and the intrinsic interpretability of simpler models such as Bayesian Knowledge Tracing (BKT). Unlike standard black-box models, iDKT utilizes a novel mechanism called

Representational Grounding to anchor its latent representations to the conceptual space of a interpretable model choosen as reference.

The core architecture of iDKT, as illustrated in Figure 1, extends the standard Transformer framework [9] by incorporating specialized components for Representational Grounding, primarily integrated within the embedding layers and the multi-objective loss pipeline.

1. **Input Data:** This stage handles the ingestion of student interactions (concepts, questions, and binary responses $r \in \{0, 1\}$). Crucially, it also loads values from the BKT reference model, including performance predictions (p_{BKT}) and per-skill parameters such as initial mastery (l_0) and learning transition rates (T), which serve as grounding targets for the model.
2. **Embeddings:** Task and history information are projected into a continuous embedding space. We then apply the individualization transformations described in Section 3 where base embeddings are augmented by student-specific parameters: a Proficiency Offset (lc) and a Rate Augmentation (tc). This produces the Transition Gap (individualized task representation x') and the Transition Gain (individualized interaction history y'), effectively defining the difficulty of a task relative to the student's baseline proficiency.
3. **Transformer Core:** The model employs an Encoder-Decoder Transformer architecture. The encoder (N blocks) processes the individualized interaction history ($y'_{1:t-1}$) through self-attention to capture global student behavior patterns. The decoder ($2N$ blocks) processes the current individualized task (x'_t) by alternating between self-attention blocks (processing the question representation) and cross-attention blocks (attending to the encoded interaction history). This cross-attention mechanism synthesizes both streams into a contextualized representation, capturing the student's current specialized knowledge state for the target task.
4. **Output Stage:** The latent Proficiency Context is passed through a multi-layer perceptron (MLP) that maps the deep representations to the final output space. The final output is the iDKT prediction (p_{iDKT}), representing the probability that the student will respond correctly to the current task.
5. **Loss Functions (Grounding Pipeline):** During training, the architecture is supervised not only by the prediction loss (L_{sup}) against ground truth outcomes but also by an alignment pipeline. This includes the Alignment Loss (L_{ref}) which penalizes deviations from the BKT prediction, and Parameter Losses (L_{init}, L_{rate}) that ground the model's internal proficiency and rate parameters to their BKT-derived theoretical counterparts.

3. Embeddings

In standard educational datasets, such as ASSISTments 2009, ASSISTments 2015, Algebra 2005, and others [28], student interactions are recorded at the level of specific questions or tasks, each of which is associated with one or more underlying concepts or knowledge components. This structure reflects the fact that proficiency in a concept (e.g., the Pythagorean Theorem) is acquired through interactions with a diverse range of tasks. While all tasks involving a concept share the same semantic core, they differ in their specific manifestations—most notably in their intrinsic difficulty or complexity. Therefore, an effective representation must capture both the shared identity of the concept and the unique deviation of the specific task.

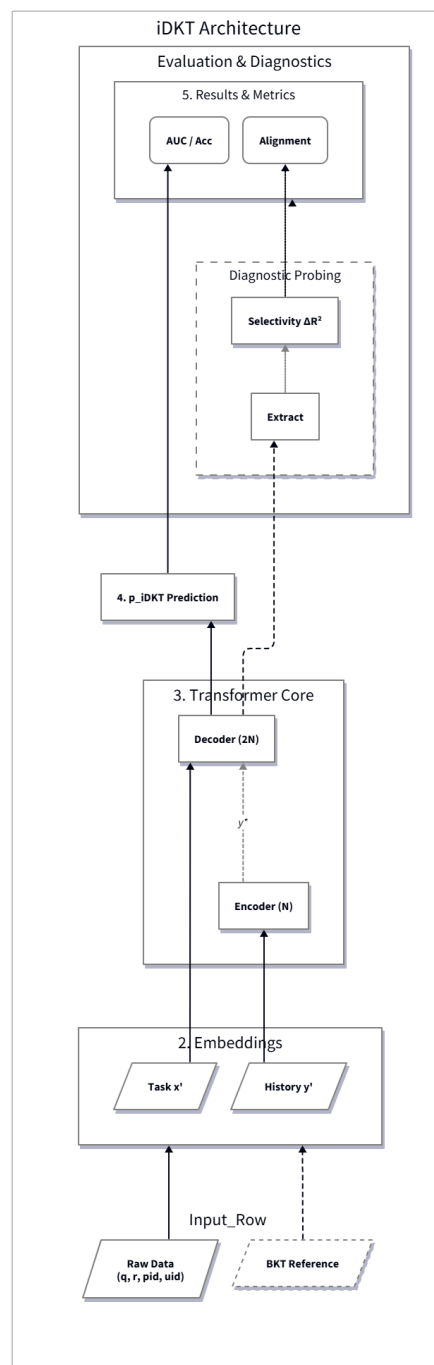


Figure 1. The iDKT Architecture. The diagram illustrates the complete pipeline: (1) **Input Layer** with BKT reference data, (2) **Representational Grounding** where input embeddings incorporate personalized initial knowledge (l_c) and learning rate (t_c) derived from BKT parameters, (3) **Encoder-Decoder Transformer** processing interaction history and tasks, (4) **Prediction** via concatenated representations and MLP, (5) **Multi-objective Loss Functions** for grounding, and (6) **Diagnostic Probing Validation** that validates preservation of pedagogical constructs through linear probes with control tasks.

In Transformer-based models [9] we can operationalize this principle representing the tasks as embedding vectors:

$$x' = c + u \cdot d \quad (\text{Task}) \quad (1)$$

In this formulation, c acts as the *Concept Anchor*, a vector representing the invariant semantic identity of the concept while the vector d represents the learnable *Question Variation Axis*, defining the specific direction of the "transition gap". The scalar u serves as the *Relational Magnitude*, representing the question's specific relative difficulty compared to other questions involving the same concept. Consequently, rather than encode arbitrary embeddings for every question, we encode the vector sum of two distinct components with clear semantic meaning: a base *concept identity* and a *difficulty shift*.

In a similar way, we can operationalize the interactions between questions and students as embedding vectors:

$$y' = e + u \cdot (f + d) \quad (\text{Interaction History}) \quad (2)$$

Here e represents the *Interaction Base*, which is a combined representation of concept c and the binary outcome r (correct/incorrect), while f represents the *Interaction Variation Axis*, which is similar to the Question Variation Axis (d_c) but is specific to the interaction between a question and a student. The inclusion of d_c in the interaction shift ensures that the difficulty vectors are consistent across both questions (x') and interactions (y').

Extending this rationale, we can enrich the x' embeddings by integrating additional components with explicit semantic significance. Specifically, by adopting Bayesian Knowledge Tracing (BKT) as a reference model, we can incorporate vectors corresponding to its core theoretical parameters—Initial Knowledge (L_0) and Learning Rate (T)—thereby grounding the deep representation in established pedagogical constructs.

To get individualized values for these parameters, we decompose them into population-level bases and student-specific deviations:

$$l_c = L_0 + k_s \cdot d_k \quad (\text{Personalized Initial Knowledge}) \quad (3)$$

$$t_c = T + v_s \cdot d_v \quad (\text{Personalized Learning Rate}) \quad (4)$$

where l_c is the personalized initial knowledge for concept, t_c is the personalized learning rate for concept, L_0 and T are the population-level base embeddings, d_k and d_v are the learnable variation axes vectors (similar to the difficulty axis d), and k_s and v_s are the scalar student-specific deviations learned for each individual.

We include these vectors to get the final input embedding for the encoder and decoder components of the Transformer:

$$x' = (c + u \cdot d) - l_c \quad (\text{Individualized Task}) \quad (5)$$

$$y' = (e + u \cdot (f + d)) + t_c \quad (\text{Individualized Interaction History}) \quad (6)$$

where c represents the concept embedding, u the question-specific difficulty shift, d the task variation axis, e the interaction base, f the interaction variation axis, l_c the personalized initial knowledge, and t_c the personalized learning rate.

The rationale for using difference for the individualized task (x') and sum for the interaction history (y') is due to their distinct semantic roles:

- x' represents the *Transition Gap: Difficulty – Proficiency*. Under this relational logic, objective task difficulty is offset by prior proficiency, ensuring that task demands are defined relative to the subject's baseline. This formulation captures the difference between the task requirements and the current state, representing the residual gap after accounting for latent proficiency.
- y' represents the *Transition Gain: Interaction + Rate*. The history encoder accumulates evidence from interactions, applying a consistent relational logic where the signal value is augmented by latent rate, ensuring that interaction outcomes are defined relative to the subject's pace. Under this formulation, the total value encompasses not only the interaction outcome but also the rate of progress through the state trajectory, as this incremental gain serves as a robust indicator of future performance. This approach, therefore, captures latent progression by augmenting observed outcomes with transition rate, thereby reflecting individualized acquisition rates.

4. Loss Functions

The model is trained using a multi-objective loss function designed to ensure that the high-capacity Transformer remains aligned with pedagogical principles through Representational Grounding. The total loss $\mathcal{L}_{total}$ is defined as a weighted sum of different loss components described in detail below.

$$\mathcal{L}_{total} = L_{sup} + \lambda_{ref}L_{ref} + \lambda_{init}L_{init} + \lambda_{rate}L_{rate} + L_{reg} \quad (7)$$

4.1. Supervised Alignment (L_{sup})

The primary objective L_{sup} uses standard Binary Cross-Entropy (BCE) between the iDKT performance predictions $\hat{y}_t$ and the observed ground truth outcomes $r_t \in \{0, 1\}$. This loss ensures predictive accuracy by minimizing the deviance from observed student behavior:

$$L_{sup} = -\frac{1}{N} \sum_{t=1}^N [r_t \log(\hat{y}_t) + (1 - r_t) \log(1 - \hat{y}_t)] \quad (8)$$

4.2. Representational Grounding

The grounding losses (L_{ref} , L_{init} , L_{rate}) use Mean Squared Error (MSE) to anchor deep representations to BKT-derived values. Specifically, L_{ref} forces behavioral predictions to stay close to the theoretical baseline, while L_{init} and L_{rate} ground the individualized parameters l_c and t_c in meaningful educational starting points and acquisition paces, respectively. Instead of arbitrary latent weights, the model's internal states are projected through a sigmoid activation $\sigma(\cdot)$ and compared directly to the reference values:

$$L_{ref} = \text{MSE}(\hat{y}, p_{BKT}) \quad (9)$$

$$L_{init} = \text{MSE}(\sigma(\bar{l}_c), L0_{BKT}) \quad (10)$$

$$L_{rate} = \text{MSE}(\sigma(\bar{t}_c), T_{BKT}) \quad (11)$$

where $\bar{l}_c$ and $\bar{t}_c$ are the average across the feature dimension of the individualized embeddings for proficiency and rate, respectively. This formulation forces the deep representation to be not only predictive but also semantically consistent with the reference constructs.

4.3. Inductive Bias Regularization (L_{reg})

While the grounding losses anchor the global position of the latent space to the BKT parameter estimations, L_{reg} ensures that student-level individualization is *parsimonious*. This loss acts directly on the individualization parameters (u_q, k_s, v_s) to ensure that the

model only deviates from the theoretical prior when functionally necessary. We apply distinct L_2 penalties to the scalar parameters governing variation:

$$L_{reg} = \lambda_u \sum_{q \in Q} u_q^2 + \lambda_k \sum_{s \in S} k_s^2 + \lambda_v \sum_{s \in S} v_s^2 \quad (12)$$

where u_q represents item difficulty, k_s is the student-specific knowledge gap, and v_s is the learning rate deviation. This formulation implements a *normal student prior*: the model assumes every subject adheres to the population-level parameters derived from the BKT reference unless their unique interaction history provides sufficient signal to justify the regularization cost.

5. Experimental Setup

5.1. Research Questions

The experimental validation of iDKT is guided by the following research questions:

1. **Interpretability Validation (RQ1):** Is it possible to rigorously validate that a iDKT model, whose representations are grounded in a reference model, actually yields interpretable constructs?
2. **Trade-Off between Predictive Performance and Interpretability (RQ2):** To what extent can deep knowledge tracing models be constrained for interpretability alignment without significantly degrading predictive performance?
3. **Improvement of Pedagogical Diagnostics (RQ3):** How can the Transformer's ability to capture longitudinal context be utilized to improve or complement pedagogical diagnostics?

5.2. Implementation Details

We implemented the iDKT model in PyTorch and used the benchmark library PYKT [28] to leverage standardized data preprocessing, dataset splitting and benchmarking of baseline models. For training, we used standard 5-fold cross-validation with an 80/20 train/test split. The model was trained using the Adam optimizer with a learning rate of $1e-4$, a batch size of 64, and a dropout rate of 0.2 to prevent overfitting. The maximum number of epochs was set to 200, with an early stopping mechanism (patience=10) to terminate training if validation performance plateaued.

The Transformer architecture was configured with an embedding dimension d_{model} of 256, 8 attention heads, and 4 encoder/decoder blocks. The feed-forward dimension d_{ff} was set to 512. Regularization penalties for individualization parameters (L_2 on u_q, k_s, v_s) were all set to $1e-5$. For reproducibility, all experiments were seeded (seed=42) and executed on NVIDIA A100 GPU infrastructure. Predictive performance was evaluated using Area Under the ROC Curve (AUC) and Accuracy (ACC), while interpretability was assessed using the metrics defined in Section 6.2.

5.3. Datasets

We did the evaluation of iDKT with these 5 widely used datasets:

- ASSISTments2009: A dataset consisting of math exercises, collected from the free online tutoring ASSISTments platform in 2009-2010. It is one of the most widely used and has been the standard benchmark for many years [29].
- ASSISTments2015: This dataset was collected from the ASSISTments platform in the year of 2015. It has the largest number of students among the other ASSISTments datasets [29].
- Algebra2005: A dataset from the KDD Cup 2010 EDM Challenge containing questions from the Carnegie Learning Algebra system deployed 2005-2006 [30].

- Bridge2006: A dataset from the KDD Cup 2010 EDM Challenge with the Carnegie Learning Bridge to Algebra system, deployed 2006-2007 [31].
- NIPS34: A dataset collected from the Eedi platform [32] containing answers to multiple-choice diagnostic math questions for the Tasks 3 & 4 at the NeurIPS 2020 Education Challenge.

6. Empirical Results

6.1. Predictive Performance

To evaluate the predictive performance of the iDKT model, we used the Area Under the Curve (AUC) and the Classification Accuracy (ACC) of the models on the test set with the 5 datasets described in Section 5.3. The results are shown in the Table 1.

Table 1. Predictive Performance of the iDKT model across 5 datasets.

Model	AS2009_S	AS2015	Algebra2005	Bridge2006	NIPS34
iDKT	0.8255	0.7252	0.9281	0.8092	0.7987

When comparing these results with state-of-the-art DKT models reported in [28], we observe that iDKT achieves the best predictive performance on the AS2009 and Algebra2005 datasets and ranks second on the Bridge2006 and NIPS34 datasets, surpassed only by the AKT model.

6.2. Interpretability Validation (RQ1)

To verify that iDKT's internal representations—specifically Personalized Initial Knowledge (I_c) and Personalized Learning Rate (t_c) as defined in Equations 3 and 4—faithfully represent the educational constructs postulated by the reference model, we employ two metrics widely utilized in psychometrics and educational measurement [33,34]:

1. *Convergent Validity (Latent Fidelity)*: This metric, denoted as I_1 , is calculated as the Pearson correlation (r) between the projected latent factors (l_u, t_u) and the reference BKT parameters (L_0 and T). High alignment proves the model has successfully internalized the theoretical constructs. The metrics are expressed as:

$$I_{1l} = \text{Corr}(l_u, L_0) \quad \text{and} \quad I_{1t} = \text{Corr}(t_u, T) \quad (13)$$

where the projected latent factors are calculated via:

$$l_u = \sigma(\bar{l}_c), \quad t_u = \sigma(\bar{t}_c) \quad (14)$$

with $\bar{l}_c$ and $\bar{t}_c$ representing the mean across the feature dimension of the individualized embeddings.

2. *Predictor Equivalence (Behavioral Alignment)*: This metric, denoted as I_2 , assesses the functional substitutability of iDKT parameters by executing a *cross-model simulation*. We "inject" the projected iDKT parameters (l_u and t_u) into the canonical BKT recurrence equations to generate *induced mastery trajectories*. This allows us to verify whether the learned factors preserve their causal roles. The metric is statistically quantified as the Pearson correlation coefficient between these induced trajectories and the theoretical reference trajectories generated by the original BKT model:

$$I_2 = \text{Corr}(P(L)_{ind}, P(L)_{ref}) \quad (15)$$

where $P(L)_{ind}$ and $P(L)_{ref}$ represent the sequences of mastery probabilities at each timestep for the induced and reference models, respectively.

Table 2 shows interpretability metrics for the unconstrained iDKT model ($\lambda_{ref} = 0$).

Table 2. Interpretability Alignment Metrics for unconstrained iDKT ($\lambda_{ref} = 0$).

Dataset	I_1 (Init.)	Int.	I_1 (Rate)	Int.	I_2	Int.
AS2009	-0.1382	Poor	-0.0067	Negl.	0.1870	Poor
AS2015	-0.0392	Negl.	0.0908	Negl.	0.0975	Negl.
Algebra2005	-0.0602	Negl.	-0.0273	Negl.	0.0600	Negl.
Bridge2006	-0.0645	Negl.	0.0160	Negl.	0.0809	Negl.
NIPS34	0.2070	Poor	-0.0425	Negl.	0.1722	Poor

Upon enabling Representational Grounding ($\lambda_{ref} = 0.10$), we observe a significant restoration of semantic alignment, as detailed in Table 3.

Table 3. Interpretability Alignment Metrics for grounded iDKT ($\lambda_{ref} = 0.1$).

Dataset	I_1 (Init.)	Int.	I_1 (Rate)	Int.	I_2	Int.
AS2009_S	0.5409	Good	0.3131	Fair	0.6250	Excell.
AS2015	0.9217	Excell.	0.8801	Excell.	0.1749	Poor
Algebra2005	0.5444	Good	0.3310	Fair	0.0661	Negl.
Bridge2006	0.4561	Fair	0.6422	Good	0.0859	Negl.

The results presented in Tables 2 and 3 reveal a notable divergence between *Convergent Validity* (I_1) and *Predictor Equivalence* (I_2). Although I_1 attains high levels across most datasets even with relatively low λ_{ref} grounding weights, I_2 remains consistently low.

High I_1 scores demonstrate that iDKT successfully internalizes the *semantic identity* of the theoretical constructs (Initial Knowledge and Learning Rate). However, I_2 measures functional substitutability within a comparatively constrained reference model that is unable to capture the intricate behavioral patterns identified by iDKT. As a high-capacity Transformer, iDKT captures complex, long-range dependencies and context-aware dynamics that exceed the modeling capability of classical BKT. This discrepancy provides empirical evidence that iDKT does not merely mimic the reference model, but rather maps its core pedagogical constructs onto a more sophisticated and predictive architecture.

6.2.1. Latent Structure Validation via Diagnostic Probing

While the correlation-based metrics I_1 and I_2 provide evidence of semantic alignment, a critical question remains: do these correlations reflect genuine structural encoding of pedagogical constructs in the model’s representations, or could they arise from superficial pattern matching? To address this concern rigorously, we employ *diagnostic probing with control tasks* [35], the gold-standard methodology for validating that deep learning representations truly encode specific theoretical constructs.

The probing protocol operates as follows. By design, iDKT’s input embeddings incorporate grounded representations of BKT constructs through Equations 3 and 4, where personalized initial knowledge (l_c) and learning rate (t_c) are explicitly embedded into the task and interaction representations (x' and y') that feed the Transformer. However, the critical question is whether the Transformer core *preserves and utilizes* this pedagogical information during its contextualization process, or whether it discards or distorts these constructs in favor of purely predictive patterns.

To validate this, we extract the representations used for final prediction—specifically, the concatenation of the decoder output with the task embedding. The decoder output is produced through a two-stage process: first, the encoder aggregates all historical interaction context ($y'_{1:t-1}$) into a compressed representation through self-attention; second,

the decoder processes the current task (x'_t) while attending to this encoded history via cross-attention, producing a task-contextualized representation that integrates both the task-specific information and the relevant historical context. We then train a simple linear classifier (linear regression probe) to predict the BKT mastery probability $P(L_t)$ directly from these task-contextualized representations, without access to any other information. The key insight is that if the Transformer has genuinely preserved and utilized the pedagogical constructs embedded in its inputs (l_c and t_c), these constructs should remain linearly recoverable from the final representations used for prediction.

However, high performance on this *true task* alone is insufficient. High-capacity models can memorize arbitrary patterns, potentially achieving spurious correlations. To rigorously distinguish genuine encoding from memorization, we implement a *control task* [35]: we train an identical probe to predict a randomly shuffled version of the BKT mastery labels. If the model has truly preserved the structural relationship between its representations and mastery, the true task should substantially outperform the control task. We quantify this via the *Selectivity* metric:

$$\text{Selectivity} = R^2_{\text{true}} - R^2_{\text{control}}$$

(16)

where R^2_{true} measures the variance explained when predicting actual BKT mastery, and R^2_{control} measures the variance explained when predicting shuffled labels. A high positive selectivity score ($\Delta R^2 > 0.5$) provides rigorous evidence that the contextualized representations structurally encode the pedagogical construct, demonstrating that the Transformer has preserved the grounded information from its inputs rather than discarding it during attention-based contextualization.

Table 4 presents the probing results for the grounded iDKT model ($\lambda_{ref} = 0.1$) on two representative datasets.

Table 4. Diagnostic Probing Results for grounded iDKT ($\lambda_{ref} = 0.1$).

Dataset	R^2_{true}	R^2_{control}	Selectivity (ΔR^2)	Interpretation
AS2009_S	0.6731	-0.0139	0.6870	Robust Structural Encoding
AS2015	0.6472	-0.0043	0.6515	Strong Alignment

The results provide compelling validation of iDKT’s interpretability. Both datasets exhibit selectivity scores exceeding 0.65, substantially above the threshold for “strong encoding” ($\Delta R^2 > 0.5$) established in the interpretability literature [35]. The near-zero (or slightly negative) control task performance confirms that the model is not simply memorizing arbitrary patterns, but has genuinely organized its post-attention representations around the pedagogical construct of mastery. Critically, these probing results validate that the grounded information embedded in the input layer (via l_c and t_c) is preserved through the Transformer’s self-attention mechanism and remains structurally encoded in the final contextualized representations. This demonstrates that the representational grounding is not merely a superficial input manipulation, but fundamentally shapes how the model processes and represents student knowledge states.

This dual validation—combining correlation-based semantic alignment (I_1, I_2) with rigorous probing selectivity—establishes that iDKT satisfies the requirements of Semantically Aligned Interpretability. The model’s representations are both semantically meaningful (aligned with BKT constructs) and structurally genuine (validated via control tasks), providing a robust foundation for pedagogically interpretable predictions.

6.3. Accuracy–Interpretability Trade-Off (RQ2)

Prediction performance and interpretability are considered as two opposing goals. Our approach to explore such trade-off is based in building a Pareto frontier, which is constructed by gradually increasing the grounding weight λ_{ref} defined in Equation ?? and plotting the resulting values of performance and interpretability.

We measure interpretability using the *Composite Alignment* $\bar{I}$ metric, defined as the arithmetic mean of the individual alignment components I_1 and I_2 defined in Equations 13 and 15:

$$\bar{I} = \frac{1}{3}(I_{1l} + I_{1t} + I_2) \quad (17)$$

Tables 5 and 7 summarize the performance and alignment results for the ASSISTments 2009 and 2015 datasets across a systematic sweep of the grounding weight λ_{ref} (Equation ??).

Figures 6 and 8 illustrate the resulting Pareto frontiers. We observe that theoretical guidance serves as a powerful regularizer, although the *Saturation Point* (the weight at which interpretability is maximized) varies by dataset complexity. For the problem-level *ASSISTments 2009*, synergy is achieved at $\lambda = 0.10$ and saturation at $\lambda = 0.30$. Conversely, the concept-level *ASSISTments 2015* reaches saturation faster at $\lambda = 0.10$, retaining 99.9% of accuracy. Beyond these points, we observe a *grounding collapse* in both datasets, where the model fails to reconcile excessive constraints with behavioral evidence.

6.4. Improvement of Pedagogical Diagnostics (RQ3)

In the following sections, we analyze how the architectural design of iDKT can be leveraged to improve pedagogical diagnostics

6.4.1. Contextualization

The capability of iDKT to leverage longitudinal context is illustrated in Figure 2, which contrasts its mastery estimations with the BKT baseline for a high-proficiency learner. While BKT leads to excessive volatility—where isolated failures are interpreted as significant knowledge drops—iDKT maintains a stable and resilient diagnostic trajectory.

The core of this advantage lies in the self-attention mechanism, which enables iDKT to encode longitudinal context by dynamically weighting the entire interaction history. Instead of relying solely on the most recent state, iDKT evaluates each interaction in the context of the student’s established behavioral patterns. In the scenario shown in Figure 2, the model maintains a high mastery estimation despite sporadic failures because the attention weights remain anchored to the student’s longitudinal progression. This capacity to distinguish between temporary behavioral noise (e.g., slips) and genuine changes in knowledge state represents a fundamental improvement over classical baselines, supporting robust and individualized diagnostics.

6.4.2. Individualization

The primary value of iDKT lies in its ability to transform population-level theoretical averages into high-granular contextual diagnostics. While standard BKT assumes that every student shares a fixed initial mastery (L_0) and learning rate (T) for a given skill, iDKT decomposes these into individualized profiles.

To statistically quantify this granularity, we analyze the *delta distribution* ($\Delta = t_s - T$), representing the student-specific deviation from the theoretical learning rate across all skills. In both ASSISTments datasets, we observe a significant non-zero standard deviation ($\sigma \approx 0.019$ for AS2009 and $\sigma \approx 0.039$ for AS2015). This represents a substantial increase in diagnostic resolution compared to the BKT baseline, where $\sigma = 0$. The right-skewed tail in

Table 5. Pareto Sweep results for ASSISTments 2015.

λ_{ref}	Performance (AUC)	Interpretability (I)	Interpretation
0.0 (Baseline)	0.7248	-0.0012	Black Box
0.1 (Saturation Point)	0.7245	0.6673	99.9% AUC Retained
0.2	0.7181	0.6453	High-Fidelity Diagnostic
0.3	0.7112	0.6095	Latent Degradation
0.4	0.7047	0.6377	Latent Degradation
0.5	0.6975	0.5691	Theory-Dominant
0.6	0.6918	0.5337	Latent Degradation
0.7	0.6874	0.5331	Latent Degradation
0.8	0.6830	0.4828	Grounding Collapse
1.0	0.6763	0.4828	Grounding Collapse

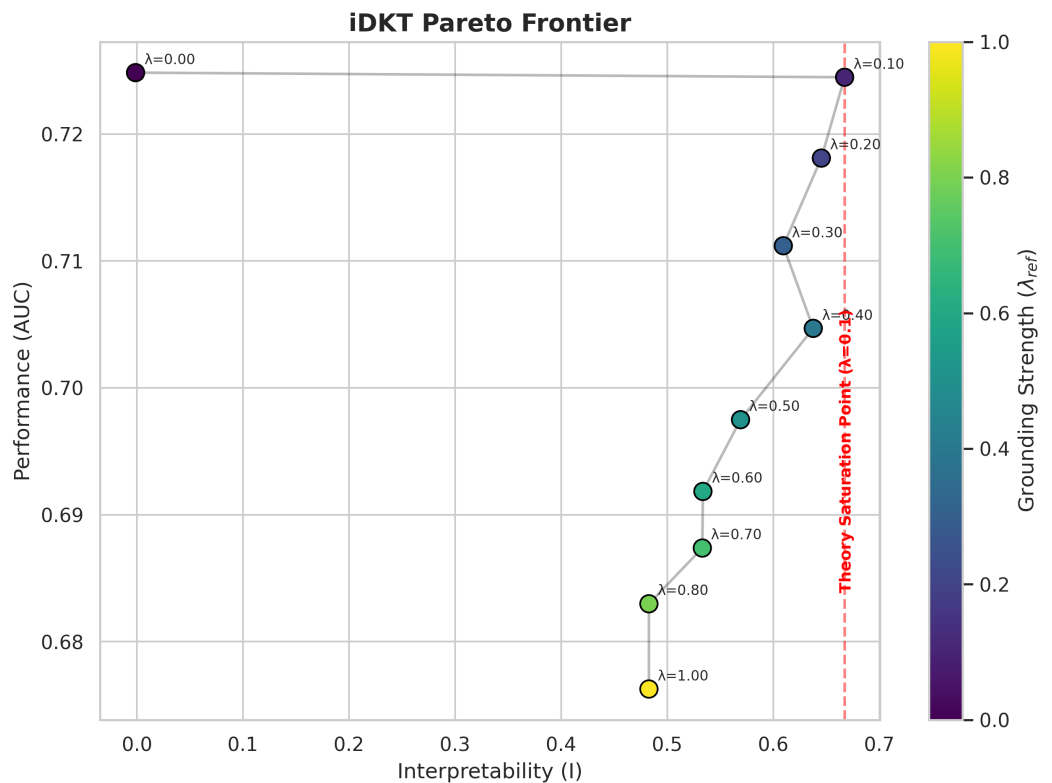


Table 6. iDKT Pareto Frontier for ASSISTments 2015. The trajectory illustrates a saturation of theoretical alignment at $\lambda \approx 0.2$, followed by a drift at higher weights where performance is traded without gaining interpretability.

Table 7. Pareto Sweep results for ASSISTments 2009.

λ_{ref}	Performance (AUC)	Interpretability (I)	Interpretation
0.0 (Baseline)	0.8207	0.0293	Unconstrained Black Box
0.1 (Synergy Spot)	0.8255	0.3846	Accuracy-Interpretability Synergy
0.2	0.8110	0.4270	Theory-Balanced Inductive Bias
0.3 (Saturation Point)	0.7963	0.4790	Theoretical Saturation (Max Int.)
0.4	0.7824	0.2740	Non-monotonicity
0.5	0.7760	0.4125	Grounding Degradation
0.6	0.7692	0.3587	Grounding Degradation
0.7	0.7629	0.2781	Grounding Degradation
0.8	0.7552	0.0708	Grounding Collapse
1.0	0.7501	0.0714	Grounding Collapse

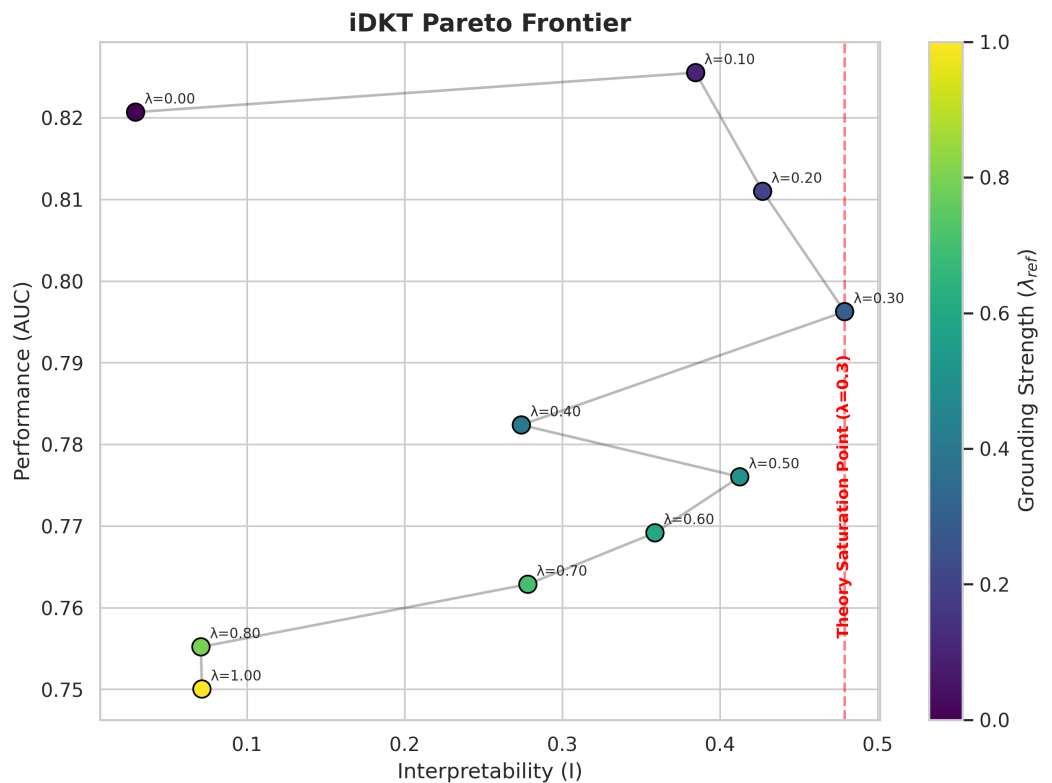


Table 8. iDKT Pareto Frontier for ASSISTments 2009. The curve displays a *synergy point* where performance peaks at $\lambda = 0.1$, followed by a *grounding collapse* as excessive constraints over-regularize the latent space.

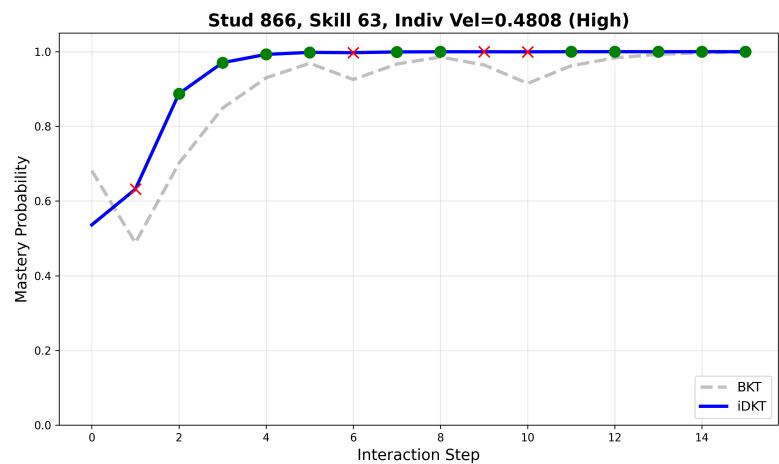


Figure 2. Comparison of two mastery trajectories estimated by iDKT and BKT illustrating how iDKT can be more stable and resilient against behavioral noise.

the distributions identifies a sub-population of fast learners whose true acquisition pace is systematically underestimated by classical models.

Figure ?? visualizes the delta distributions for both datasets. The width of the curves quantifies the pedagogical information that is lost when using population-level averages. This variance allows the system to distinguish between students who are faster than suggested by theoretical priors and those who require additional practice to reach mastery.

6.4.3. High-Granularity Trajectory Analysis

To provide definitive visual evidence for RQ3, we generated a Mastery Alignment Mosaic (Figure 3) displaying individualized trajectories for three learning archetypes (Fast, Median, Slow) across 15 skills. For rigorous validation, we utilized the “Set S Isolation” strategy, restricting the analysis to students with continuous interaction histories (≤ 200 steps) and recalibrating the BKT reference exclusively on this population.

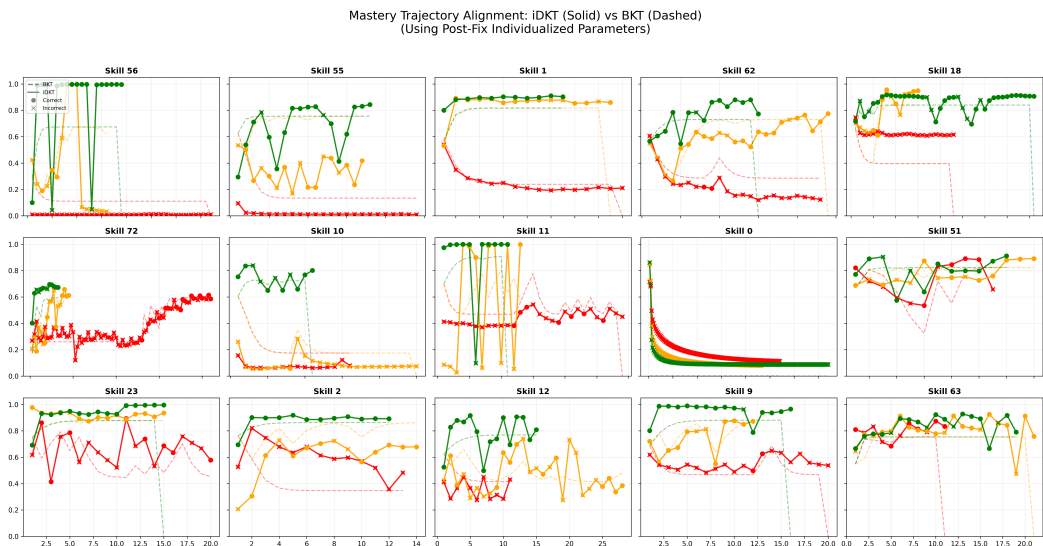


Figure 3. Mastery Trajectory Alignment Mosaic for AS2009_S. Solid lines represent iDKT individualized mastery, while dashed lines show the BKT baseline. Markers indicate observed student performance (Circle: Correct, X: Incorrect). Trajectories are selected from students with > 10 interactions per skill.

The qualitative analysis of these trajectories reveals three significant pedagogical insights:

1. **Dynamic Diagnostic Placement:** Even when BKT is forced to start at the population prior (L_0), iDKT starting points ($t = 1$) vary per student. This demonstrates that iDKT leverages cross-skill behavioral transfers to perform individualized placement before the first interaction with a new skill.
2. **Trajectory Crossovers:** We observe instances where students starting with lower initial mastery overtake peers due to a higher identified learning velocity (t_s). This confirms iDKT's ability to model individualized acquisition rates beyond population-level curves.
3. **Diagnostic Resilience:** iDKT displays superior stability compared to BKT baselines. While isolated failures cause sharp drops in BKT dashed lines, iDKT solid lines for high-velocity learners remain stable, correctly identifying temporary slips vs. fundamental knowledge loss.

6.4.4. Confidence

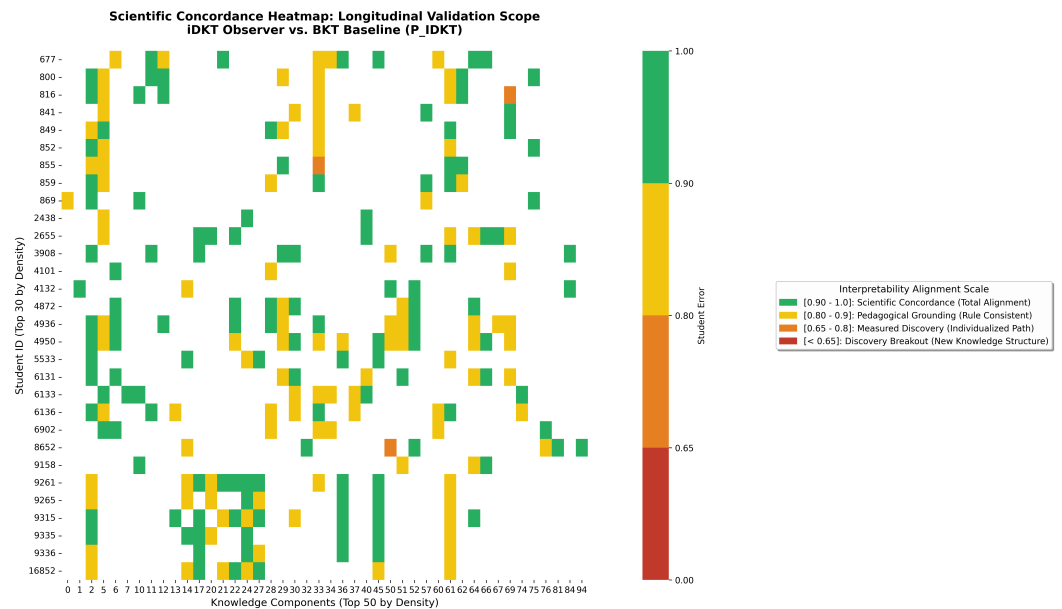
Individualized student profiling offers new opportunities for pedagogical monitoring and adaptive intervention. A significant practical application is evaluating the reliability of predictions made by intrinsically interpretable models, such as BKT. By comparing BKT mastery estimations with those from iDKT, we can derive a measure of pedagogical confidence in the theoretical baseline.

This relationship is visualized in Figure 4, which displays the concordance between BKT and iDKT mastery predictions across the sub-population of the most frequent skills and students for both datasets. The ASSISTments 2015 results demonstrate high levels of agreement, whereas the ASSISTments 2009 data reveals more divergence, where iDKT uncovers individualized patterns beyond the theoretical prior. Such visualizations provide actionable pedagogical insights that complement traditional baseline diagnostics. This visualization framework provides robust support for personalized longitudinal tracking by identifying students or curricular content that deviate from theoretical expectations. These divergences serve as critical diagnostic indicators for cases that may warrant prioritized pedagogical intervention.

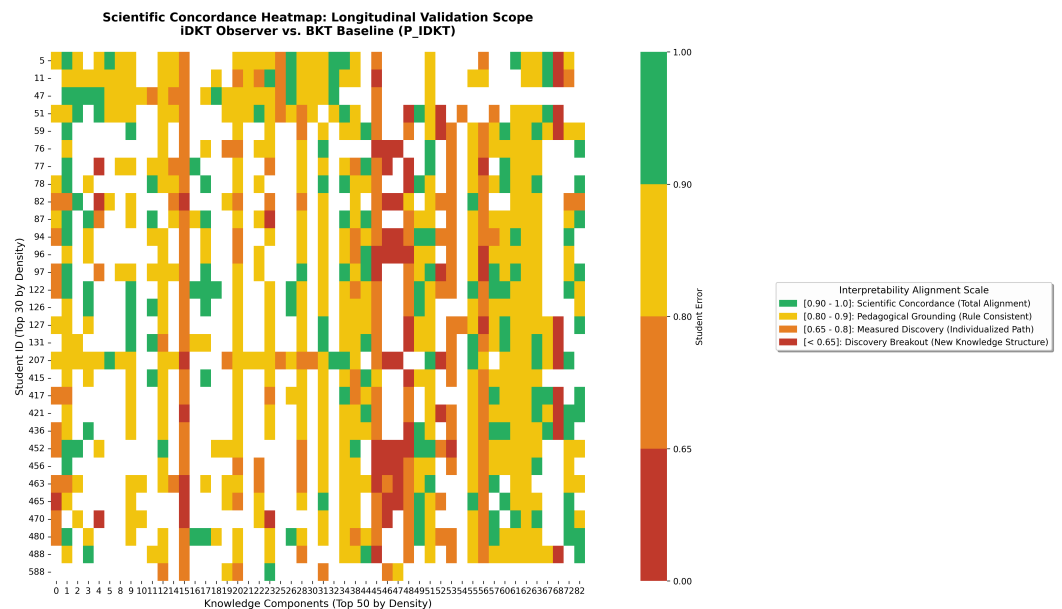
7. Conclusions

This work has introduced iDKT, a Transformer-based Knowledge Tracing model that bridges the gap between the high predictive capacity of deep learning and the intrinsic interpretability of models such as BKT, which utilize constructs with clear semantic meanings. By utilizing Representational Grounding, we have demonstrated that it is possible to anchor deep latent representations to semantically meaningful constructs without sacrificing state-of-the-art performance.

Our experimental results successfully address the research questions. First, we proved that iDKT successfully internalizes the theoretical constructs of the reference model, achieving high convergent validity (I_1) and demonstrating robust structural encoding through diagnostic probing with control tasks (selectivity $\Delta R^2 > 0.65$). This dual validation—combining correlation-based alignment with rigorous probing selectivity—establishes that the semantic alignment reflects genuine structural encoding rather than superficial pattern matching. Second, we formalized a methodology for measuring the trade-off between predictive performance and interpretability, enabling the identification of the “Interpretability Sweet Spot” where significant alignment is achieved with minimal loss in predictive AUC. Finally, we demonstrated that iDKT provides a high increase in di-



(a) ASSISTments 2015



(b) ASSISTments 2009

Figure 4. Heatmaps of per-skill prediction alignment.

agnostic granularity compared to population-level baselines by identifying individualized learning velocities that enable truly adaptive pacing.

The primary contribution of this research is twofold: a robust methodology for evaluating the internal interpretability of Transformer-based models in education, and a practical architecture that transforms “black-box” predictors into interpretable tools for knowledge tracing. By grounding deep learning in established educational concepts, iDKT offers a path toward AI-driven personalization that is both highly accurate and pedagogically actionable, providing a foundation for next-generation intelligent tutoring systems.

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